Concentric Zero-Spheres of a class of sections of the commutative ring $\mathcal{R}$ of slice preserving functions over the compactified Octonions

with the Riemann Hypothesis as a corollary

Jesse Michael Paul
PhD Candidate, Department of Mathematics & Statistics
University of North Carolina at Greensboro

June 2026

"... a 'viewpoint' by itself remains fragmentary. It reveals to us one of the aspects of a scenery or panorama, among a multiplicity of others which are equally valuable, equally 'real'. It is when complementary viewpoints of a common reality are conjugated, that is, when our 'eyes' are multiplied, that the gaze is able to penetrate further ahead in the reckoning of things. ... it also happens, sometimes, that a sheaf of viewpoints converging to a unique and vast scenery gives rise to a novel thing; a thing which transcends each of the partial perspectives, in the same way that a living being transcends each of its limbs and organs. This new thing could be called a vision."

— A. Grothendieck, *Récoltes et Semailles*

Abstract

Let $\mathcal{R}$ be the commutative ring of slice-preserving slice-regular functions on the compactified octonions $\mathbb{O}^* = S^8$, under the regular $(*)$ product. Part 1 gathers the classical facts about the Riemann zeta function. Part 2 develops the slice-preserving theory, builds $\mathcal{R}$, places the octonionic zeta $\zeta_{\mathbb{O}^*}$ inside it, and proves the zero and equivalence theorems: the classically nontrivial zeros of ζ are residue-$\mathbb{C}$ zero 6-spheres, and the Riemann Hypothesis holds exactly when those spheres are concentric, sharing a single real centre.

Part 3 proves the main result, the *Concentricity Theorem*, for a section $A \in \mathcal{R}$ satisfying four conditions: (C1) meromorphic continuation with a single simple pole carried to $\infty = N$; (C2) an infinite Euler product; (C3) an infinite slice-regular Weierstrass factorization whose spherical factors are the residue-$\mathbb{C}$ zeros; and (C4) infinitely many such zeros. The theorem says the residue-$\mathbb{C}$ zero 6-spheres of any such A are concentric.

The proof builds a concentricity detector from the section itself, out of groupoids, orbit–stabilizer, and categorical homotopy theory. Conditions (C1)–(C3) determine one distinguished Möbius element on the projective base $\mathcal{B} = PGL(2,\mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$: the Euler product presenting it at 0, the Weierstrass product at N, through the commuting triangle $\pi \circ E = \exp$. The slice-preserving normalization is G_2-equivariant on $\mathbb{O}^*$, and orbit–stabilizer carries this whole point-valued action across the base. Its sphere-world projection records the directions and Möbius legs; the genuine A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$ has the resulting normalized A-value states and their transports as its fibres. Its Grothendieck construction is the total $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$; the residue-$\mathbb{C}$ zero-spheres enter as its (C3)–(C4) outputs. The readout $\pi_0(\mathcal{T}_A) \cong \mathrm{colim}_{\mathcal{B}}(\pi_0 \circ \mathcal{A}_A)$ collapses the section's real-value transports to a single component carrying one real value $c \in \mathbb{R}$, so every populated zero 6-sphere has centre c. The theorem asserts a single shared centre c; naming that number is the task of the corollaries.

Downstream corollaries translate residue-$\mathbb{C}$ zeros to classically nontrivial zeros and, for $\zeta_{\mathbb{O}^*}$, use its functional equation to identify $c = \frac{1}{2}$: the Riemann Hypothesis.

Contents

Introduction

The Riemann Hypothesis is usually met head-on, as a question about where the zeros of a single complex function fall along a line. This paper approaches it from the side. It asks instead whether a family of geometric objects, sitting a few dimensions up in the octonions, share a common centre, and shows that they must. The Riemann Hypothesis then follows, not as the thing we set out to prove, but as the shadow this larger fact casts back down onto the complex plane.

A nontrivial zero of the zeta function, seen from the complex line alone, is a lone point in a strip. Seen from the compactified octonions, that same zero opens into a round six-dimensional sphere, and the critical-line question becomes a question of arrangement: are these spheres concentric, gathered about one shared real centre? It takes the whole continuum of imaginary directions in the octonions, read together, to see whether the alignment holds.

The paper is built to make that convergence visible and, at every step, checkable. Part 1 recalls the classical facts about the Riemann zeta function we will lean on. Part 2 develops the slice-preserving theory of functions on the octonions, builds the commutative ring $\mathcal{R}$ in which the octonionic zeta lives, and proves the reframing at the centre of everything: the classically nontrivial zeros are exactly the residue-$\mathbb{C}$ zero six-spheres, and the Riemann Hypothesis holds precisely when those spheres are concentric. Part 3 proves the main result, the Concentricity Theorem, with a small piece of machinery I think of as a concentricity detector. From the section itself we build one distinguished action on a projective base, carry it across a continuum of Riemann spheres by orbit and stabilizer, and read off, through categorical homotopy theory, whether the zero-spheres land in a single connected piece. They do; and for the octonionic zeta, that is the Riemann Hypothesis.

No one can picture an eight-dimensional graph in their head, but tools from categorical homotopy theory, like the total-object Grothendieck construction and standard colimit arguments, let us probe this structure and reason about it. The argument is also formalized in Lean, so that each step can be checked; the reader can not only verify the steps but, I hope, come to understand for themselves why the Concentricity Theorem is true and why the Riemann Hypothesis follows.

Part I
Classical Background

1 The Riemann zeta function

The Riemann zeta function is classical, and we take its basic theory as given. We recall only what the construction uses: the meromorphic continuation and functional equation, the Euler product over the primes, the Hadamard product over the zeros, and Hardy's theorem. We then read ζ in compactified form, as a map to the Riemann sphere, which is the form in which it will extend to the octonions.

Theorem 1.1 (Meromorphic continuation and functional equation; Riemann [13]). *The Dirichlet series $\zeta(s) = \sum_{n\geq 1} n^{-s}$, convergent for $\operatorname{Re} s > 1$, continues to a meromorphic function on $\mathbb{C}$ with a single simple pole, at $s = 1$; equivalently, a holomorphic map to the Riemann sphere*

$$\zeta : \mathbb{C}^* \longrightarrow \mathbb{C}^*, \qquad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \qquad \zeta(1) = \infty.$$

The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire and satisfies

$$\xi(s) = \xi(1-s), \qquad \xi(\bar{s}) = \overline{\xi(s)}.$$

Consequently, if ρ is a nontrivial zero of ζ then so are $\bar{\rho}$, $1-\rho$, and $1-\bar{\rho}$. The trivial zeros of ζ are at $s = -2, -4, \dots$; the nontrivial zeros lie in the strip $0 < \operatorname{Re} s < 1$. The Riemann Hypothesis asserts that every nontrivial zero has $\operatorname{Re} s = \frac{1}{2}$.

Theorem 1.2 (Infinite Euler product; [15, Ch. 1]). *For $\operatorname{Re} s > 1$,*

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 - p^{-s}\right)^{-1} = \exp\Big(\sum_{p} \sum_{k\geq 1} \tfrac{1}{k}\, p^{-ks}\Big),$$

the product taken over the infinitely many primes; it converges absolutely and is zero-free on $\operatorname{Re} s > 1$.

Theorem 1.3 (Infinite Hadamard product; [15, Ch. 2]). *ξ is entire of order 1 with infinitely many zeros, which are exactly the nontrivial zeros $\{\rho\}$ of ζ, and admits the Hadamard factorization*

$$\xi(s) = \xi(0) \prod_{\rho} \Big(1 - \tfrac{s}{\rho}\Big) e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

Corollary 1.4 (Infinitude of the nontrivial zeros). *ζ has infinitely many nontrivial zeros.*

Proof. Immediate from Theorem 1.3: its zeros are infinitely many and are exactly the nontrivial zeros of ζ. □

Theorem 1.5 (Hardy [11]). *Infinitely many nontrivial zeros of ζ lie on the line $\operatorname{Re} s = \frac{1}{2}$.*

The compactified Riemann zeta on $\mathbb{C}^*$

For the slice construction, ζ is taken in compactified form, with values on the Riemann sphere $\mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2$ (the one-point compactification of $\mathbb{C}$ [12]). The classical facts above are retained as stated.

Definition 1.6 (Compactified classical zeta). $\zeta_{\mathbb{C}^*} : \mathbb{C}^* \to \mathbb{C}^*$ is the meromorphic continuation of ζ regarded as a map to the Riemann sphere: $\zeta_{\mathbb{C}^*}(s) = \zeta(s)$ for $s \in \mathbb{C} \setminus \{1\}$; $\zeta_{\mathbb{C}^*}(1) = \infty$ (the simple pole, Theorem 1.1); and $\zeta_{\mathbb{C}^*}(\infty) = 1$, since $\zeta(s) \to 1$ as $\operatorname{Re}(s) \to +\infty$ (the p-test: only the $n = 1$ term of $\sum_n n^{-s}$ survives). It is holomorphic into $\mathbb{C}^*$ on $\mathbb{C}^* \setminus \{1\}$, its sole pole being $s = 1$. The functional equation, the conjugate symmetry $\zeta_{\mathbb{C}^*}(\bar{s}) = \overline{\zeta_{\mathbb{C}^*}(s)}$, and Hardy's infinitude (Theorem 1.5) hold verbatim for $\zeta_{\mathbb{C}^*}$, as they concern values on $\mathbb{C} \setminus \{1\}$ where $\zeta_{\mathbb{C}^*} = \zeta$.

Lemma 1.7 (Compactification does not move the zeros). *$Z(\zeta_{\mathbb{C}^*}) = Z(\zeta)$: the zeros of $\zeta_{\mathbb{C}^*}$ in $\mathbb{C}^*$ are exactly the zeros of ζ in $\mathbb{C}$; in particular the nontrivial zeros coincide.*

Proof. On $\mathbb{C} \setminus \{1\}$ the two functions agree, so they share all zeros there. The two adjoined values are not zeros: $\zeta_{\mathbb{C}^*}(1) = \infty$ (a pole) and $\zeta_{\mathbb{C}^*}(\infty) = 1$; and $s = 1$, being a pole of ζ, was never a zero. Hence no zero is gained or lost. □

Part II
Slice-Preserving Theory, the Ring $\mathcal{R}$, and the Equivalence

2 Octonions and slices

The octonions are an 8-dimensional real division algebra, the largest of the normed division algebras. This section fixes what we need from them: the algebra itself; Artin's theorem, that any two octonions generate an associative subalgebra, so that powers and slicewise complex analysis make sense; and the complex slices $\mathbb{C}_v$, one for each imaginary direction $v \in S^6$, together with their compactifications, the slice Riemann spheres. All the slices share the real axis and a single point at infinity, the north pole N; that shared geometry is what the later construction turns around.

The octonions $\mathbb{O}$ have basis $\{1, e_1, \ldots, e_7\}$ with each $e_i^2 = -1$.

Definition 2.1. The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*: $|ab| = |a||b|$; (iii) *non-associative*: $(ab)c \neq a(bc)$ in general; (iv) *alternative*: $(aa)b = a(ab)$ and $a(bb) = (ab)b$.

An octonion $s \in \mathbb{O}$ decomposes as $s = \sigma + w$ with $\sigma = \mathrm{Re}(s) \in \mathbb{R}$ and $w = \mathrm{Im}(s) \in \mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$; the unit imaginary octonions form $S^6 \subset \mathrm{Im}(\mathbb{O})$.

Theorem 2.2 (Artin [14]). *In an alternative algebra, the subalgebra generated by any two elements is associative.*

Corollary 2.3. *Powers s^n are well-defined for any $s \in \mathbb{O}$.*

Definition 2.4 (Complex slices and slice Riemann spheres). For any unit imaginary octonion $v \in S^6$, the *complex slice* is $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$. Since $v^2 = -1$, this is a subalgebra isomorphic to $\mathbb{C}$ via the $\mathbb{R}$-algebra isomorphism $\varphi_v : \mathbb{C} \to \mathbb{C}_v$, $i \mapsto v$. All slices share the real axis: $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$ for $v \neq \pm w$. Its one-point compactification is the *slice Riemann sphere* $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$, and φ_v extends to an isomorphism of Riemann spheres $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ with $\varphi_v(\infty) = \infty$ (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

3 Slice-preserving functions

The class we use is the *slice-preserving* slice-regular functions. We first recall slice regularity [8, 6, 5, 9], holomorphicity on each complex slice separately, as the ambient notion, then isolate the slice-preserving subclass, on which classical complex methods apply slicewise (by Artin's theorem each slice $\mathbb{C}_v$ is an associative subalgebra where classical complex analysis applies unchanged).

Definition 3.1 (Slice regularity; [5, Def. 5.1.1]). Let $\Omega \subseteq \mathbb{O}$ be open. A function $f : \Omega \to \mathbb{O}$ is *slice regular* if for each $v \in S^6$, $\varphi_v^{-1} \circ f \circ \varphi_v$ satisfies the Cauchy–Riemann equations on $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$. A domain Ω is *axially symmetric* if $\sigma + \gamma v \in \Omega$ implies $\sigma + \gamma w \in \Omega$ for all $w \in S^6$.

Theorem 3.2 (Representation Formula; [5, Thm. 5.1.7]). *Let f be slice regular on an axially symmetric slice domain Ω. Fix $v_0 \in S^6$ and write $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$ with $\alpha, \beta : \mathbb{R}^2 \to \mathbb{R}$. Then for every $v \in S^6$, $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$, where $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$ and $\beta = \frac{1}{2} v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$.*

Theorem 3.3 (Identity Theorem; [5, Cor. 5.1.9]). *Let f be slice regular on an axially symmetric slice domain Ω. If f vanishes on a subset of a single slice $\Omega \cap \mathbb{C}_{v_0}$ with an accumulation point in $\Omega \cap \mathbb{C}_{v_0}$, then $f \equiv 0$ on all of Ω.*

Theorem 3.4 (Extension Theorem; [5, Thm. 5.1.5]). *Let $U \subseteq \mathbb{C}$ be a domain symmetric under conjugation and $g : U \to \mathbb{C}$ holomorphic. There is a unique slice regular $\mathrm{ext}(g) : \Omega_U \to \mathbb{O}$ on $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U,\ v \in S^6\}$ with $\mathrm{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$ for every v_0.*

Definition 3.5 (Slice-preserving functions; [1, Def. 2.7, Rem. 2.8]; [17, 7, 16]). A slice-regular function f on an axially symmetric domain Ω is *slice preserving* if it maps every slice into itself: $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$ for all $v \in S^6$. For octonions ($\#S^6 > 2$) the following are equivalent characterisations:

(i) f is *intrinsic*: $f(q^c) = f(q)^c$ for all $q \in \Omega$, where q^c is octonionic conjugation (on a slice, $a + bv \mapsto a - bv$);

(ii) its representation-formula components are $\mathbb{R}$-valued: writing $f(\sigma+\gamma v)=\alpha(\sigma,\gamma)+v\,\beta(\sigma,\gamma)$ (Theorem 3.2), one has $\alpha,\beta:\mathbb{R}^2\to\mathbb{R}$, equivalently the inducing stem function $F=F_1+\iota F_2$ has $\mathbb{R}$-valued components F_1,F_2;

(iii) f preserves two distinct slices; equivalently $f(\Omega\cap\mathbb{R})\subseteq\mathbb{R}$ when Ω is a slice domain.

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of $\mathcal{R}$ (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti– Stoppato [7, Rem. 1.8]).

Remark 3.6 (Compactified extension). Definition 2.4 through Definition 3.5, and Theorems 3.2–3.4, are stated in the literature on uncompactified domains $\Omega\subseteq\mathbb{O}$ (and $U\subseteq\mathbb{C}$). We use them on the compactified $\mathbb{O}^*=S^8$ and on the slice Riemann spheres $\mathbb{C}_v^*=S_v^2$ (GPV's slice Riemann manifolds [17]). The passage $\mathbb{O}\rightsquigarrow\mathbb{O}^*$, $\mathbb{C}_v\rightsquigarrow\mathbb{C}_v^*$ adjoins only the shared point $\infty=N$; it is recorded as a marked extension node (in the Lean development: cite the uncompactified statement and extend through the one-point compactification `OnePoint`).

4 The octonionic zeta function $\zeta_{\mathbb{O}^*}$

Slice-preserving theory already gives the tools to build the octonionic zeta function. On each slice $\mathbb{C}_v$ we read the compactified classical ζ through the isomorphism $\varphi_v:\mathbb{C}^*\to\mathbb{C}_v^*$, and the representation formula glues these slice copies into a single function on all of $\mathbb{O}^*$. The one thing to check is that the two identifications of a slice with $\mathbb{C}$ agree, so that $\zeta_{\mathbb{O}^*}$ is well-defined; they do, because conjugating the input and reversing the identification cancel. The pole of ζ at $s=1$ is carried to the north pole N.

Definition 4.1 (Octonionic zeta)**.** Here N denotes the *north pole* of S^8: the single point at infinity ∞ adjoined in the one-point compactification $\mathbb{O}^*=\mathbb{O}\cup\{\infty\}\cong S^8$ (inverse stereographic projection; GPV [17]). It is one point with two names, so $\infty=N$; it is the only point at infinity, shared by $\mathbb{O}^*$ and by every slice Riemann sphere $\mathbb{C}_v^*=S_v^2$.

The *octonionic zeta function* $\zeta_{\mathbb{O}^*}:\mathbb{O}^*\to\mathbb{O}^*$ is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications $\varphi_v:\mathbb{C}^*\xrightarrow{\sim}\mathbb{C}_v^*=S_v^2$ (the $\mathbb{R}$-algebra isomorphism $\mathbb{C}\to\mathbb{C}_v$ of Definition 2.4, extended to the Riemann spheres by $\varphi_v(\infty)=N$; GPV [17], Prop. 4.1/Thm. 4.2):

(i) for $s\in\mathbb{O}\setminus\mathbb{R}$, with $w=\mathrm{Im}(s)$, $v=w/|w|\in S^6$, $\gamma=|w|>0$, $\sigma=\mathrm{Re}(s)$, so $s=\sigma+\gamma v$: $\zeta_{\mathbb{O}^*}(s):=\varphi_v(\zeta_{\mathbb{C}^*}(\sigma+i\gamma))$;

(ii) for $s\in\mathbb{R}\setminus\{1\}$: $\zeta_{\mathbb{O}^*}(s):=\zeta_{\mathbb{C}^*}(s)=\zeta(s)\in\mathbb{R}$;

(iii) for $s=1$ (the simple pole, Definition 1.6): $\zeta_{\mathbb{O}^*}(1):=\infty=N$;

(iv) for $s=\infty=N$: $\zeta_{\mathbb{O}^*}(\infty):=\zeta_{\mathbb{C}^*}(\infty)=1$.

Proposition 4.2 (Well-definedness)**.** *$\zeta_{\mathbb{O}^*}:\mathbb{O}^*\to\mathbb{O}^*$ is well-defined.*

Proof. For $s\in\mathbb{O}\setminus\mathbb{R}$ the decomposition $s=\sigma+\gamma v$ with $\gamma>0$, $v\in S^6$ is forced ($\gamma=|\mathrm{Im}\,s|$, $v=\mathrm{Im}\,s/|\mathrm{Im}\,s|$), so the only ambiguity is the identification of the slice $\mathbb{C}_v=\mathbb{C}_{-v}$ with $\mathbb{C}^*$. That slice carries two such identifications, φ_v ($i\mapsto v$) and φ_{-v} ($i\mapsto -v$), under which s reads as $\sigma+i\gamma$ resp. $\sigma-i\gamma$; we must check $\varphi_v(\zeta_{\mathbb{C}^*}(\sigma+i\gamma))=\varphi_{-v}(\zeta_{\mathbb{C}^*}(\sigma-i\gamma))$. By conjugate symmetry

$\zeta_{\mathbb{C}^*}(\sigma - i\gamma) = \overline{\zeta_{\mathbb{C}^*}(\sigma + i\gamma)}$ ([15, Ch. 2]; Definition 1.6), writing $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = a + bi$ gives $\varphi_{-v}(a - bi) = a + (-b)(-v) = a + bv = \varphi_v(a + bi)$: the two sign reversals, one from conjugating the input, one from the reversed identification, cancel. The remaining cases carry no ambiguity: for real $s \neq 1$, $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$ for every v; at $s = 1$ the value is the single point $\infty = N$; and at $s = \infty = N$ the value is $\zeta_{\mathbb{C}^*}(\infty) = 1 \in \mathbb{R}$, again the same for every v since each φ_v fixes $\mathbb{R}$ pointwise. □

5 $\mathcal{R}$ is a commutative ring

The slice-preserving sections form a commutative ring $\mathcal{R}$, with $\zeta_{\mathbb{O}^*}$ among its elements. Its multiplication is the regular ($*$) product, which descends from the Cayley–Dickson construction of $\mathbb{O}$ and from slice regularity; on slice-preserving functions it restricts on each slice to the ordinary pointwise product (Wang), and is therefore commutative. The ring axioms, and the integral-domain property, follow slicewise.

We work on $\mathbb{O}^*$; write $\Omega_v := \mathbb{O}^* \cap \mathbb{C}_v = \mathbb{C}_v^* = S_v^2$ for the slice Riemann sphere (Definition 2.4).

Definition 5.1 (The ring of slice-preserving functions under the regular product)**.** $\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$ is the set of all slice-regular functions $f : \mathbb{O}^* \to \mathbb{O}^*$ that are slice preserving (Definition 3.5), $f(\Omega_v) \subseteq \mathbb{C}_v^*$ for all $v \in S^6$, equipped with pointwise addition and the regular ($*$) product.

Commutativity of $\mathcal{R}$ rests on a single slice-preserving fact from the literature: on each slice the regular product is ordinary multiplication.

Theorem 5.2 (Regular product on slice-preserving functions; [16, Rem. 2.11])**.** *For slice-preserving f, g (with $f(\Omega_v) \subseteq \mathbb{C}_v$), the regular product restricts on each slice to the pointwise product, $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$, and $f * g = g * f$, $(f * g) * h = f * (g * h)$ (via Artin, Theorem 2.2).*

Proposition 5.3. *$(\mathcal{R}, +, *)$ is a commutative ring with identity $1 \in \mathcal{R}$.*

Proof. Closure under $+$ is pointwise. For $*$: fixing a slice $\mathbb{C}_K$, the product reduces to pointwise multiplication on $\mathbb{C}_K$ by Theorem 5.2, which is commutative because $\mathbb{C}_K$ is a commutative associative subalgebra; associativity and distributivity are inherited slicewise, and a slice-regular function is determined by its restriction to any one slice (Theorem 3.3). The constant 1 is the identity. □

Proposition 5.4 ($\mathcal{R}$ is an integral domain)**.** *If $f, g \in \mathcal{R}$ and $f * g = 0$, then $f = 0$ or $g = 0$.*

Proof. $(f * g)|_{\Omega_K}(w) = f(w) \cdot g(w)$ in $\mathbb{C}_K \cong \mathbb{C}$ (Theorem 5.2). The restrictions $f|_{\Omega_K}$, $g|_{\Omega_K}$ are holomorphic on the connected slice; by the classical identity principle the pointwise product vanishes identically iff one factor does. By Theorem 3.3 the corresponding global function vanishes. □

6 The slice-preserving analytic toolkit on the compactified octonions

We collect the slice-preserving analytic facts used in Part 3 as cited statements: the exponential, its logarithm manifold, its degenerate fibre, the slice/stem square and I-independent lift, and the winding lift. Each is established in the literature on the uncompactified $\mathbb{O}$ and used here on $\mathbb{O}^* = S^8$ and the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (the passage is recorded in Remark 3.6; the compactified slice-manifold structure itself is GPV's [17]).

Theorem 6.1 (The slice exponential; [21, §3]; [17, Prop. 5.1]; [1, Rem. 2.23])**.** *The exponential* $\exp : \mathbb{O} \to \mathbb{O} \setminus \{0\}$, $\exp(q) = \sum_{k \ge 0} q^k/k!$ *(powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by*

$$\exp(x + Iy) = e^x(\cos y + I \sin y), \qquad x, y \in \mathbb{R},\ I \in S^6.$$

Consequently $|\exp q| = e^{\operatorname{Re} q}$ *depends only on the real part, and* exp *is nowhere zero.*

Theorem 6.2 (The logarithm manifold $E^+_{\mathbb{O}}$; [17, Prop. 5.1, Rem. 5.2, Def. 5.3, Prop. 5.4, Def. 5.5]; [21, §3])**.** *Let* $T : \mathbb{O} \to \mathbb{O} \times \operatorname{Im} \mathbb{O}$, $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y,\ Iy)$*, and* $\mathbb{O}^+ = \{q : \operatorname{Re} q > 0\}$*. The* logarithm manifold *is* $E^+_{\mathbb{O}} := T(\mathbb{O}^+)$*, the semi-helicoidal hypercomplex* 8*-manifold; equivalently it is the image of the exponential graph, parametrised by the* $E^+_{\mathbb{O}}$-exponential

$$E : \mathbb{O} \longrightarrow E^+_{\mathbb{O}} \subset \mathbb{O} \times \operatorname{Im} \mathbb{O}, \qquad E(x + Iy) = (\exp(x + Iy),\ Iy) = (e^x \cos y + Ie^x \sin y,\ Iy),$$

an immersion and a diffeomorphism, with inverse the $E^+_{\mathbb{O}}$-logarithm $L : E^+_{\mathbb{O}} \to \mathbb{O}$, $L(q, p) = \log |q| + p$ *(here* $p = \operatorname{Arg}(q)$*). Writing* π *for the projection to the first factor,* $\pi \circ E = \exp$ *(Theorem 6.1); but* $\pi : E^+_{\mathbb{O}} \to \mathbb{O} \setminus \{0\}$ *is* not *a covering and not an open map, precisely because* exp *is not open. Thus* $E^+_{\mathbb{O}}$ *is an adapted blow-up of* $\mathbb{O}$ *along the real axis: it unwraps the multivalued logarithm into the single-valued* L*, and a branch of the hypercomplex logarithm is* $\log_{\mathbb{O}} = L \circ (\pi|_{\Omega})^{-1}$ *on any path-connected* $\Omega \subset E^+_{\mathbb{O}}$ *on which* $\pi|_{\Omega}$ *is injective.*

Lemma 6.3 (The degenerate set of the exponential; its fibre over a real value)**.** Sourced. *With* π *the first-factor projection,* $\pi \circ E = \exp$ ([17, Rem. 5.2(a)], the commuting triangle)*; and, unlike the complex setting,* $\pi : E^+_{\mathbb{O}} \to \mathbb{O}$ *is not a covering and not an open map, because* exp *is not open: it has a non-empty* degenerate set consisting of spheres ([17, Rem. 5.2(b)], verbatim)*. The direction* $I(q)$ *admits no continuous extension to any point of* $\mathbb{R}$ ([21, Rem. 2.1])*.*

Proved from the slice form. *For* $r > 0$*,*

$$\exp^{-1}(-r) = \big\{ \log r + I(2k{+}1)\pi :\ I \in S^6,\ k \in \mathbb{Z} \big\} :$$

by Theorem 6.1, $\exp(x + Iy) = e^x \cos y + Ie^x \sin y = -r$ *forces* $e^x \sin y = 0$*, hence* $y \in \pi\mathbb{Z}$*; negativity of the value forces* $y = (2k{+}1)\pi$ *and* $x = \log r$*; the direction* I *is unconstrained. The fibre is thus indexed by the single real* level $\log r = \log |-r|$ *and, within it, by the odd winding indices* $2k{+}1$ *carried as band data (the winding lift, Theorem 6.5; [21, Cor. 5.21]). The fibre formula itself appears, stated without proof as Preface motivation, in [17] (p. 972); the slice-form derivation above is the load-bearing statement.*

Definition 6.4 (Slice restriction and the I-independent lift; [16, Rem. 2.11]; [4, §3.3, §3.7]; [7, Def. 2.1, Prop. 2.2])**.** A slice-preserving section A carries each slice Riemann sphere into itself,

$$A(\mathbb{C}^*_v) \subseteq \mathbb{C}^*_v = S^2_v \qquad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v :\ \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}^*_v$ of Definition 2.4. The *I-independent lift* is the fact that this holomorphic stem is the *same* for every v: there is one holomorphic $A_\circ : \mathbb{C}^* \to \mathbb{C}^*$ with

$$A|_{\mathbb{C}^*_v} = \varphi_v \circ A_\circ \circ \varphi_v^{-1} \qquad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently, A is *intrinsic*, $A(q^c) = A(q)^c$, and is induced ($A = \mathcal{I}(F)$) by a single stem function $F = F_1 + \iota F_2 : D \to \mathbb{O}_{\mathbb{C}}$ on the symmetric $D \subset \mathbb{C}$ whose components F_1, F_2 are $\mathbb{R}$-valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I\, F_2(x + iy), \qquad F_1, F_2\ \mathbb{R}\text{-valued}.$$

It is this single I-independent slice stem that makes a slice-preserving section G_2-equivariant, hence blind to the G_2-orbit direction (Definition 9.2).

Theorem 6.5 (The winding lift; [21, Def. 3.4, Def. 4.1, Prop. 4.2, Def. 5.11])**.** *Let $\gamma : [a, b] \to \mathbb{O}\setminus\{0\}$ be a path (on each slice $\mathbb{C}_v$ this is the classical complex continuation). A* continuation *of the* hypercomplex logarithm *along γ is a path $\tilde\gamma : [a, b] \to \mathbb{O}$ with $\exp\circ\tilde\gamma = \gamma$, and a* lift *of γ to the logarithm manifold $E^+_{\mathbb{O}}$ (Theorem 6.2) is a path $\Gamma : [a, b] \to E^+_{\mathbb{O}}$ with $\mathrm{pr}_1 \circ \Gamma = \gamma$:*

$$\begin{tikzcd} & \mathbb{O} \\ {[a,b]} \arrow[ur, "\tilde\gamma"] \arrow[r, "\gamma"'] & \mathbb{O}\setminus\{0\} \arrow[u, "\exp"'] \end{tikzcd} \qquad \begin{tikzcd} & E^+_{\mathbb{O}} \arrow[d, "\mathrm{pr}_1"] \\ {[a,b]} \arrow[ur, "\Gamma"] \arrow[r, "\gamma"'] & \mathbb{O}\setminus\{0\} \end{tikzcd}$$

The two data are equivalent*: a lift Γ exists if and only if a continuation $\tilde\gamma$ exists, and they correspond by*

$$\tilde\gamma = L \circ \Gamma, \qquad \Gamma = (\exp\circ\tilde\gamma,\ \mathrm{Im}\,\tilde\gamma),$$

where L is the $E^+_{\mathbb{O}}$-logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a loop *$\gamma : S^1 \to \mathbb{O}\setminus\{0\}$ the lift is taken on the universal cover $\exp : i\mathbb{R} \to S^1$, as a continuous $\Gamma : i\mathbb{R} \to E^+_{\mathbb{O}}$ with $\mathrm{pr}_1 \circ \Gamma = \gamma \circ \exp$ ([21, Def. 5.11]):*

$$\begin{tikzcd} i\mathbb{R} \arrow[r, "\Gamma"] \arrow[d, "\exp"'] & E^+_{\mathbb{O}} \arrow[d, "\mathrm{pr}_1"] \\ S^1 \arrow[r, "\gamma"'] & \mathbb{O}\setminus\{0\} \end{tikzcd}$$

Proposition 6.6 (Tameness, existence, and the winding number; [21, Def. 4.20, Cor. 5.21, Cor. 5.22])**.** *The lift of Theorem 6.5 is governed by the companion structure of γ.*

(i) Uniqueness (tameness). *γ is* tame *when it admits a* unique *companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).*

(ii) Existence and closure. *A lift of a loop γ exists if and only if the signature satisfies $\sigma(\gamma|_{[\xi_l,\xi_{l+1}]}) \in \{0, -1\}$ on each obstruction interval; and* if it exists, the lift is itself a loop *([21, Cor. 5.13]; pinned against the arXiv text; the earlier citation to Cor. 5.22 is superseded).*

(iii) Winding number. *For an untwisted tame loop with even circular signature σ^c, the winding number is $\omega = |\sigma^c|/2$ ([21, Cor. 5.21]).*

7 Slice preservation and symmetries

With the ring in hand, we check that $\zeta_{\mathbb{O}^*}$ actually lives in it, and record the symmetry that will organize its zeros. Slice preservation says $\zeta_{\mathbb{O}^*}$ keeps each slice inside itself; together with slice regularity this makes it a section of $\mathcal{R}$. The symmetry is the action of the automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$, which fixes the real axis and rotates the imaginary directions S^6, and $\zeta_{\mathbb{O}^*}$ is equivariant under it. This is what forces the non-real zeros to come in whole 6-spheres rather than isolated points, the geometry the next section reads off.

Theorem 7.1 (Slice preservation). *$\zeta_{\mathbb{O}^*}$ is slice preserving (Definition 3.5): $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ for every $v \in S^6$.*

Proof. Fix $v \in S^6$ and $s \in \mathbb{C}_v^*$. If $s \in \mathbb{C}_v \setminus \mathbb{R}$ then $s = \sigma \pm \gamma v$ and $\zeta_{\mathbb{O}^*}(s) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma \pm i\gamma)) \in \varphi_v(\mathbb{C}^*) = \mathbb{C}_v^*$ (Definition 4.1(i)). If $s \in \mathbb{R}\setminus\{1\}$ then $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$; while $\zeta_{\mathbb{O}^*}(1) = \infty \in \mathbb{C}_v^*$ and $\zeta_{\mathbb{O}^*}(\infty) = 1 \in \mathbb{C}_v^*$. In every case the value remains in $\mathbb{C}_v^*$. $\square$

Theorem 7.2 ($\zeta_{\mathbb{O}^*}$ is a section of $\mathcal{R}$). *$\zeta_{\mathbb{O}^*} \in \mathcal{R}$.*

Proof. On each slice the restriction is the compactified classical zeta read through the isomorphism φ_v, namely $\zeta_{\mathbb{O}^*}|_{\mathbb{C}_v^*} = \varphi_v \circ \zeta_{\mathbb{C}^*} \circ \varphi_v^{-1}$, holomorphic into $\mathbb{C}_v^*$ away from the single pole. Equivalently $\zeta_{\mathbb{O}^*} = \mathrm{ext}(\zeta)$ is the slice-regular extension of the holomorphic, conjugation-symmetric ζ on $\mathbb{C} \setminus \{1\}$ (Theorem 3.4, [5, Thm. 5.1.5]); hence $\zeta_{\mathbb{O}^*}$ is slice regular on $\mathbb{O}^* \setminus \{1, \infty\}$ (slice regularity is slicewise holomorphy, Definition 3.1). By Theorem 7.1 it is slice preserving, $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$. A slice-regular, slice-preserving function $\mathbb{O}^* \to \mathbb{O}^*$ is by definition a section of $\mathcal{R} = \mathcal{O}_{\mathbb{O}^*}$ (Definition 5.1); the regular product on such sections restricts on each slice to the ordinary product (Theorem 5.2, Wang [16, Rem. 2.11]). Finally $\zeta_{\mathbb{O}^*}$ is not a unit, since it has zeros (Theorem 8.2). $\square$

Definition 7.3 (The automorphism group G_2). $G_2 = \mathrm{Aut}(\mathbb{O})$ is the 14-dimensional compact, connected, simply connected Lie group of $\mathbb{R}$-algebra automorphisms of $\mathbb{O}$. Each $g \in G_2$ fixes 1, hence the real axis pointwise, and acts on $\mathrm{Im}\,\mathbb{O} \cong \mathbb{R}^7$ as an element of $SO(7)$; thus $g \cdot (\sigma + \gamma v) = \sigma + \gamma\, g(v)$.

Theorem 7.4 (G_2-equivariance). *$\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$ for all $g \in G_2$, $s \in \mathbb{O}$.*

Proof. For $s \in \mathbb{R}$, $g \cdot s = s$ and $\zeta_{\mathbb{O}^*}(s) \in \mathbb{R}$ is fixed, so both sides equal $\zeta_{\mathbb{O}^*}(s)$. For a non-real $s = \sigma + \gamma v$ ($v \in S^6$, $\gamma > 0$), g carries the slice $\mathbb{C}_v$ to the slice $\mathbb{C}_{g(v)}$ by the isometric relabelling $\varphi_{g(v)} = g \circ \varphi_v$ (Definition 7.3). Since $\zeta_{\mathbb{O}^*}$ is defined slicewise by the same $\zeta_{\mathbb{C}^*}$ read through φ_v (Definition 4.1(i)),

$$\zeta_{\mathbb{O}^*}(g \cdot s) = \varphi_{g(v)}\big(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)\big) = g\big(\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))\big) = g \cdot \zeta_{\mathbb{O}^*}(s).$$

$\square$

Theorem 7.5 ([3]). *G_2 acts transitively on S^6 with stabilizer $\mathrm{SU}(3)$: $\mathrm{SU}(3) \hookrightarrow G_2 \twoheadrightarrow S^6$.*

Remark 7.6 (The G_2-action extends to $\mathbb{O}^* = S^8$). The action above is stated on $\mathbb{O}$, but $\mathcal{H}_1$ (Part 3) runs on the compactified $\mathbb{O}^* = S^8$. Each $g \in G_2$ fixes the real axis pointwise (an automorphism fixes 1, hence all of $\mathbb{R}$) and acts on $\mathrm{Im}\,\mathbb{O} = \mathbb{R}^7$ as an element of $SO(7)$; in particular g is a linear isometry of $\mathbb{R}^8 = \mathbb{O}$, hence proper, and therefore extends uniquely to a homeomorphism of the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ fixing the point at infinity $\infty = N$ (the one-point compactification is functorial on proper maps; in Lean, `OnePoint`). Thus $G_2 \curvearrowright \mathbb{O}^*$ fixes the compactified real axis $\mathbb{R} \cup \{\infty\}$, a great circle through N, pointwise, and acts on each direction-sphere S^6 exactly as in Theorem 7.5. This is the action on which $\mathcal{H}_1$ is built, with $\mathcal{R}$ its ring of invariants over $\mathbb{O}^*$.

8 Zero geometry and the equivalence

This section is the bridge the whole paper is built to cross. It shows that the zeros of $\zeta_{\mathbb{O}^*}$ are exactly the classical nontrivial zeros, now spread out: each nontrivial zero ρ of ζ becomes a whole 6-sphere of zeros of $\zeta_{\mathbb{O}^*}$, its residue-$\mathbb{C}$ zero-sphere, sitting at real part $\operatorname{Re}\rho$. The Riemann Hypothesis, that every ρ has real part $\frac{1}{2}$, becomes the statement that all these 6-spheres share the same real part, that they are concentric. From here on we work with the spheres.

Theorem 8.1 (Zero Equivalence Theorem). *Let $\rho = \sigma + i\gamma \in \mathbb{C}^*$. For every $v \in S^6$,*

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \quad \Longleftrightarrow \quad \zeta_{\mathbb{C}^*}(\rho) = 0.$$

Equivalently: a non-real $s \in \mathbb{O}^*$ is a zero of $\zeta_{\mathbb{O}^*}$ if and only if its slice coordinate $\varphi_v^{-1}(s)$ is a zero of $\zeta_{\mathbb{C}^*}$; by Lemma 1.7 these are exactly the nontrivial zeros of the classical ζ.

Proof. By Definition 4.1(i), $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$. The slice identification $\varphi_v : \mathbb{C}^* \to \mathbb{C}_v^* = S_v^2$ is an isomorphism of slice Riemann spheres (the $\mathbb{R}$-algebra isomorphism $\mathbb{C} \to \mathbb{C}_v$ of Definition 2.4, extended by $\varphi_v(\infty) = N$; GPV [17]). An isomorphism is injective and sends 0 to 0, so $\varphi_v(z) = 0 \iff z = 0$; with $z = \zeta_{\mathbb{C}^*}(\rho)$ this gives both implications. (Slice preservation, Theorem 7.1, is what places the value in $\mathbb{C}_v^*$ so that φ_v^{-1} applies on the non-real side.) The final clause is Lemma 1.7. ☐

Theorem 8.2 (Nontrivial zeros are infinitely many disjoint 6-spheres). *For a nontrivial zero $\rho = \sigma + i\gamma$ of $\zeta_{\mathbb{C}^*}$ (so $\gamma > 0$) set*

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma\, S^6 \subset \mathbb{O}.$$

Then:

- *(i) S_ρ* is *a 6-sphere, the round sphere of radius γ centred at the real point σ, equivalently the G_2-orbit of any one of its points $\sigma + \gamma v_0$ (Theorem 7.4; G_2 acts transitively on S^6 with stabiliser* SU(3)*, Theorem 7.5, Baez [3]).*
- *(ii) every point of S_ρ is a zero of $\zeta_{\mathbb{O}^*}$, and conjugate zeros give the same sphere, $S_\rho = S_{\bar\rho}$;*
- *(iii) the nontrivial-zero set is the disjoint union $Z^{\mathrm{nt}}_{\mathbb{O}^*} = \bigsqcup_{[\rho]} S_\rho$ over conjugate pairs $[\rho] = \{\rho, \bar\rho\}$, with distinct pairs giving disjoint spheres;*
- *(iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).*

Proof. (i) The map $v \mapsto \sigma + \gamma v$ is the affine map $x \mapsto \sigma + \gamma x$ of $\operatorname{Im}\mathbb{O} \cong \mathbb{R}^7$ restricted to S^6; with $\gamma > 0$ its image is the round 6-sphere of radius γ centred at σ. By G_2-equivariance (Theorem 7.4) and transitivity of G_2 on S^6 (Theorem 7.5), $G_2 \cdot (\sigma + \gamma v_0) = \{\sigma + \gamma\, g(v_0) : g \in G_2\} = S_\rho$. So S_ρ *is* S^6, scaled by γ, translated to σ, realized as a single G_2-orbit; no embedding or diffeomorphism is invoked.

(ii) For any $v \in S^6$, $\zeta_{\mathbb{O}^*}(\sigma+\gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\rho)) = \varphi_v(0) = 0$ by the Zero Equivalence Theorem 8.1. And $S_{\bar\rho} = \{\sigma + \gamma(-v) : v \in S^6\} = S_\rho$, since $v \mapsto -v$ is a bijection of S^6.

(iii) For "$\subseteq$": a non-real zero $s = \sigma + \gamma v$ of $\zeta_{\mathbb{O}^*}$ forces $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = 0$ by Theorem 8.1, so $s \in S_\rho$ with $\rho = \sigma + i\gamma$ nontrivial (Lemma 1.7); "$\supseteq$" is (ii). Disjointness: if $\sigma + \gamma v = \sigma' + \gamma' w$, uniqueness of the real/imaginary parts gives $\sigma = \sigma'$ and $\gamma v = \gamma' w$, whence $\gamma = \gamma'$ (taking norms, $|v| = |w| = 1$) and $\{\rho', \bar\rho'\} = \{\rho, \bar\rho\}$; so two spheres coincide or are disjoint.

(iv) There are infinitely many spheres because ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). ☐

Theorem 8.3 (The equivalence: concentricity ⇔ RH)**.** *The following are equivalent:*

(a) the Riemann Hypothesis;

(b) the nontrivial-zero 6*-spheres* $\{S_\rho\}$ *of* $\zeta_{\mathbb{O}^*}$ *(Theorem 8.2) are* concentric*: they share a single common centre, necessarily a point of the real axis.*

When they hold, the common centre is $\frac{1}{2}$*.*

Proof. Each S_ρ is centred at the real point σ_ρ, where $\rho = \sigma_\rho + i\gamma_\rho$ (Theorem 8.2(i)).

$(a) \Rightarrow (b)$: under RH every $\sigma_\rho = \frac{1}{2}$, so all spheres share the centre $\frac{1}{2}$.

$(b) \Rightarrow (a)$: suppose the spheres share a common centre $c \in \mathbb{R}$. By the functional equation (Theorem 1.1), if ρ is a nontrivial zero then so is $1 - \bar{\rho} = (1 - \sigma_\rho) + i\gamma_\rho$, whose sphere $S_{1-\bar{\rho}}$ is centred at $1 - \sigma_\rho$. Concentricity forces $\sigma_\rho = c = 1 - \sigma_\rho$, hence $\sigma_\rho = \frac{1}{2}$; as ρ was arbitrary this is RH, and the common centre is $\frac{1}{2}$. □

Part III
The Categorical Construction and the Concentricity Theorem

Part 2 revealed that ζ carries a special geometry: spread onto the octonions, its zeros are concentric exactly when the Riemann Hypothesis holds. That raises the question this part answers. What general properties of a section of the ring $\mathcal{R}$ of slice-preserving functions force its residue-$\mathbb{C}$ zeros to be concentric? Following that question leads to four conditions, C1–C4, and to the whole class of functions satisfying them, which we call the A-sections. The main result, the Concentricity Theorem, is that the residue-$\mathbb{C}$ zeros of any A-section are concentric. The octonionic zeta is one A-section among many, and the Riemann Hypothesis is one of its corollaries.

From the section's own data we read one distinguished Möbius transformation of the Riemann sphere, presented at its two fixed points by the Euler product and the Weierstrass product; orbit–stabilizer carries that single transformation across the whole continuum of slice directions. The resulting map to SphereWorld is its geometric projection; lifting the same action to the normalized A-value states assembles the genuine A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$ over the projective base $\mathcal{B}$. Its Grothendieck construction $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ gathers every sphere and every value-transport into one object, and the components of that object, computed as a colimit, collapse the transports that carry the zeros down to a single class. That one class is one real centre, and the zero-spheres are concentric.

9 The geometric worlds and the categorical construction

This section assembles the parts the detector runs on. We begin with the compactified octonions $\mathbb{O}^* = S^8$ and their north pole N, the point at infinity where a section sends its pole. We then describe what a slice-preserving section does on each slice, one holomorphic stem repeated in every direction, which is why its non-real zeros fill whole 6-spheres. We set up the two geometric registers that carry this data, the octonionic world $\mathcal{H}_1$ and the slice-value world $\mathcal{S}_2$. And we close with the projective base $\mathcal{B}$, the A-section functor over it, and its Grothendieck total $\mathcal{T}_A$, the object the next section reads.

The compactified octonions and the point at infinity

Definition 9.1 (The compactified octonions $\mathbb{O}^* = S^8$)**.** Let $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ be the Alexandroff one-point compactification of $\mathbb{O} = \mathbb{R}^8$. Concretely (GPV [17], Prop. 4.1 and Thm. 4.2, for $m = \dim_{\mathbb{R}} K \in \{4, 8\}$): the inverse stereographic projections from the north and south poles,

$$f(x+Iy) = \left(\tfrac{2(x+Iy)}{1+x^2+y^2},\ \tfrac{-1+x^2+y^2}{1+x^2+y^2}\right), \qquad h(x+Iy) = \left(\tfrac{2(x-Iy)}{1+x^2+y^2},\ \tfrac{1-x^2-y^2}{1+x^2+y^2}\right),$$

give a conformal atlas $\{(K, f), (K, h)\}$ endowing S^m with the structure of a slice (quaternionic/octonionic) Riemann manifold, with transition map $h^{-1} \circ f(q) = \bar{q}/q^2 = 1/q$, itself slice regular and slice preserving. The adjoined point is the north pole N; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.Compactification.OnePoint`, $\mathbb{O}^* =$ `OnePoint` $\mathbb{O}$ with $\infty =$ `OnePoint.infty`, and the homeomorphism $\mathbb{O}^* \cong S^8$ is `onePointEquivSphereOfFinrankEq` applied to $\dim_{\mathbb{R}} \mathbb{O} = 8$ (`Mathlib.Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of $\mathbb{O}^*$, the point at infinity $\infty = N$. The geometric registers $\mathcal{H}_1$ and $\mathcal{S}_2$ are built on these compactified values (Definition 9.3); the distinct projective base $\mathcal{B}$ and direction-indexed sphere world used in the construction below are introduced in Definition 9.4.

There is one geometric north pole $N = \infty$, represented in two categorical registers. In $\mathcal{H}_1$ this point of $\mathbb{O}^*$ is a G_2-fixed object: its own connected component, with $\mathrm{Aut}(N) = G_2$, carrying no universal property. In the slice-value world $\mathcal{S}_2$ the same geometric point is represented by the object

$$\mathfrak{n} \ := \ \infty = N \text{ regarded as an object of } \mathcal{S}_2,$$

the common point at infinity of the slice Riemann spheres, fixed by band and direction alike. The two representatives belong to *different* categorical registers and are related by the section through C1: the pole of A has value ∞, so the section carries the pole point to $\mathfrak{n}$. The residue field of a closed zero sorts the picture: residue-$\mathbb{R}$ (the G_2-fixed locus $\mathbb{R} \cup \{N\}$, classically trivial) versus residue-$\mathbb{C}$ (the 6-sphere orbits, classically nontrivial; the dictionary is Part 2 and Theorem 8.2). The later total construction uses this one shared compactified witness N, the same for every zero index.

What a slice-preserving section does

Definition 9.2 (The section's slice data and equivariance)**.** Let $A \in \mathcal{R}$. By the I-independent lift (Definition 6.4; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with $\mathbb{R}$-valued components,

$$A(x+Iy) = F_1(x+iy) + I\,F_2(x+iy), \qquad F_1, F_2\ \mathbb{R}\text{-valued},$$

the same for every $I \in S^6$. Three consequences used below:

(i) *Slice preservation of values.* $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$ for every I: the value at a point lies on that point's own slice sphere (Definition 5.1).

(ii) *G_2-equivariance.* For $g \in G_2$ and non-real $q = x + Iy$,

$$A(g \cdot q) = F_1 + g(I)\,F_2 = g \cdot \big(F_1 + I\,F_2\big) = g \cdot A(q),$$

since $g \cdot (x + Iy) = x + g(I)y$ (Definition 7.3) and F_1, F_2 are real; for $q \in \mathbb{R} \cup \{N\}$ both sides are G_2-fixed, because $A(\mathbb{R}) \subseteq \mathbb{R}$ and $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$ by (i). (For $\zeta_{\mathbb{O}^*}$ this is Theorem 7.4; the computation above is the same one, run for an arbitrary intrinsic section.)

(iii) *Blindness to the sphere direction.* On a G_2-orbit $S_{(\sigma,\gamma)}$ the stem coordinates $(F_1, F_2)(\sigma + i\gamma)$ are constant; in particular if A vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

A slice-preserving section carries one holomorphic stem per slice, and the direction S^6 is traversed only by the G_2-morphisms already present in both worlds.

Condition C4, infinitely many residue-$\mathbb{C}$ zeros, matches the class to its infinite structure (the infinite Euler family of C2, the infinite Weierstrass divisor of C3) and gives the collapse its force: the degenerate fibre of the transport is an infinite family, so that whether it lies in one connected component of $\mathcal{T}_A$ (Definition 9.4) is an infinite alignment, decided by Theorem 10.6.

Two geometric viewpoints

Definition 9.3 (The octonionic and slice-value worlds)**.** Recall the *translation groupoid* of an action $G \curvearrowright X$: its objects are the points of X and its morphisms $x \to y$ are the elements $g \in G$ satisfying $g \cdot x = y$. Thus $\pi_0(G \ltimes X) = X/G$ (Quillen [22, §1]; in Lean, `CategoryTheory.ActionCategory`).

The octonionic world is

$$\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*.$$

Its components are the G_2-orbits: the compactified real fixed locus $\mathbb{R} \cup \{N\}$ and, for each $\sigma \in \mathbb{R}$ and $\gamma > 0$, the residue-$\mathbb{C}$ sphere $S_{(\sigma,\gamma)} = \{\sigma + \gamma v : v \in S^6\}$, labelled by its real centre σ and radius γ (Baez [3]; Theorem 7.5).

The slice-value world $\mathcal{S}_2$ organizes the same compactified values by their slice decomposition. Every slice contains the real axis, and

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \qquad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

Its objects are the points of $\mathbb{O}^*$. Its generating paths are the direction arrows induced by G_2 and the band phases $z \mapsto e^{I\theta} z$ within a compactified slice; the Lean object `S2` is the quotient path category by the direction-composition relations. In particular 0 and N are shared values, not new objects indexed by a zero.

These two geometric registers hold the octonionic and slice-value geometry drawn on below.

The base of the degenerate set and the Grothendieck construction

Definition 9.4 (The projective base, genuine section functor, and total category)**.** The base is the projective action groupoid

$$\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R}).$$

Its objects are points of the compactified real projective line and its arrows are projective transformations together with their transport equations.

The projective action is read in the direction-indexed sphere-world groupoid. Its objects are directions $I \in S^6$ equipped with their compactified slice spheres, and its morphisms carry the direction and Möbius legs. A normalized value state pairs such a direction with a point of its slice.

Conditions C1–C3 determine one distinguished diagonal Möbius element, the Euler product presenting it at 0 and the Weierstrass product at N, its two fixed boundary points, through the commuting triangle $\pi \circ E = \exp$. That triangle makes the element an action, and orbit–stabilizer carries it across the base: the Möbius group law gives the local action, orbit–stabilizer extends the element over every orbit, stabilizer compatibility makes the result independent of representatives, and the group identity and multiplication give the functor laws. Its direction-and-Möbius projection is

$$A^{\text{slice}} : \mathcal{B} \longrightarrow \text{SphereWorld},$$

written `AsectionSlice A` in the Lean source. This projection is not the full A-section functor: it remembers the slice sphere and its Möbius transport, but not yet the normalized point and value produced on that sphere.

The slice-preserving realization on $\mathbb{O}^*$ is G_2-equivariant, hence a functor on the octonionic action groupoid. Sweeping that whole point-valued action—rather than only its sphere projection—by the same orbit–stabilizer factorization gives the genuine A-section functor

$$\mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd},$$

written `AsectionFunctor A` in the Lean source. The winding, lift-uniqueness, tameness, crossing, welding, and GPV laws are readings of this one action. The objects of $\mathcal{A}_A(b)$ are normalized analytic and sphere-world value states produced by the action, and its arrows are the actual value transports. C1 supplies continuation through the unique simple real pole of value $N = \infty$; C2 the Euler exponential channel on its right half-space; C3 the Weierstrass channel over the full divisor; and C4 the infinite residue-$\mathbb{C}$ population. From the C2 and C3 exponential channels, slice-preserving theory yields the unique continuous winding lift each transport carries. The real-value content is thus part of the A-section states and transports before the total object is formed.

The total value-transport category is

$$\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A.$$

An object is a base position together with a normalized A-section value state in its fibre. A morphism is a base channel together with the compatible A-section transport supplied by A. The populated residue-$\mathbb{C}$ zero-sphere states are the degenerate-fibre outputs of this completed construction, produced by C3–C4. The associated diagram $\pi_0 \circ \mathcal{A}_A : \mathcal{B} \to \mathbf{Set}$ is the components diagram required by the Grothendieck/colimit readout (Lemma 10.3).

10 The concentricity theorem

We now have the worlds and the base, so this section defines the class of functions precisely and proves the theorem. Two facts about slice-preserving functions come first: the Weierstrass factorization that gives condition C3 its meaning, and the fact that a residue-$\mathbb{C}$ zero is a whole 6-sphere, a single connected component of the octonionic world. Then the categorical engine, Riehl's computation of the components of a Grothendieck construction as a colimit, which sorts the assembled total into the classes linked by the section's real-value transports. With these in hand we define an A-section by its four conditions C1–C4, state and prove the Concentricity Theorem, and record the corollary that carries its conclusion back into the language of classical zeros.

Proposition 10.1 (Slice-regular Weierstrass factorization; the content of C3)**.** *Let $A \in \mathcal{R}$ have a zero of order $m \geq 0$ at the origin and no non-real isolated zeros (for $A \in \mathcal{R}$ the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on $\mathbb{O}^* \setminus \{N\}$ it factors as*

$$A = q^m \, R \, S \, h,$$

where q denotes the octonionic coordinate function (the identity section, so q^m carries the order-m zero at the origin), R is a slice-preserving factor vanishing exactly at the real zeros of A, S is a *factor vanishing exactly at its spherical (residue-$\mathbb{C}$) zero-spheres, and h is never vanishing; each of R, S is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds* including the case where either family of zeros is infinite.

Proof. This is the factorization theorem of Altavilla–de Fabritiis [2, Prop. 3.1, Thm. 3.2], whose primary factors are the slice-regular ones of Gentili–Vignozzi [10]; convergence of the infinite products is part of the cited statement. The pair R, S is the residue split named in C3: R carries the residue-$\mathbb{R}$ (classically trivial) zeros and S the residue-$\mathbb{C}$ (classically nontrivial) ones, the latter infinite in number by C4. □

Lemma 10.2 (Residue-$\mathbb{C}$ zeros are 6-sphere components of $\mathcal{H}_1$)**.** *For $A \in \mathcal{R}$, the non-real zeros of A are 6-spheres, each a single connected component of the translation groupoid $\mathcal{H}_1$.*

Proof. A is slice-preserving, so on a sphere $\sigma + \gamma S^6$ ($\gamma > 0$) it has the form $A(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\,\beta(\sigma, \gamma)$ with α, β real (the real stem, [16]); since v is an imaginary unit, $\alpha + v\beta = 0$ forces $\alpha = \beta = 0$, independently of v. Hence a non-real zero occurs on a *whole* sphere $\sigma + \gamma S^6$ or at none of its points (the spherical-zero structure, [1]); real zeros are isolated real points (residue field $\mathbb{R}$). Each such sphere $\sigma + \gamma S^6$ is the G_2-orbit of any of its points (Theorem 7.5: G_2 acts transitively on S^6 with stabiliser SU(3)), and the connected components of $\mathcal{H}_1$ are exactly the G_2-orbits: in a groupoid two objects share a component iff a morphism joins them, and a morphism $x \to y$ of $\mathcal{H}_1$ is a $g \in G_2$ with $g \cdot x = y$. Therefore the residue-$\mathbb{C}$ zeros, the non-real ones, are exactly the 6-sphere components of $A^{-1}(0)$. □

Lemma 10.3 (π_0 of a Grothendieck construction)**.** *For a functor $F : \mathcal{B} \to$ Grpd, the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:*

$$\pi_0\Big(\int_{\mathcal{B}} F\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}}\,(\pi_0 \circ F).$$

Proof. By Riehl's proof of Lemma 8.3.4, for any $X : \mathcal{C} \to \mathbf{Set}$ each arrow of the category of elements imposes exactly the condition that the corresponding elements be identified in every cone, so that $\pi_0(\mathrm{el}\,X) \cong \mathrm{colim}_{\mathcal{C}}\,X$ [23, p. 102]. Taking $X = \pi_0 \circ F$, the colimit performs exactly the identifications generated by the maps of F. In the Lean development this equivalence is `pi0GrothendieckEquiv`, built from the canonical cocone `pi0Cocone` and `colimit.desc` over Mathlib's `colimit_sound` and `colimit_eq`. □

Definition 10.4 (A-sections)**.** An *A-section* is a section A of the ring $\mathcal{R}$ of slice-preserving slice-regular functions on $\mathbb{O}^* = S^8$ (Definition 5.1) with the following four properties.

- (C1) *Meromorphic continuation; single simple pole.* A is meromorphically continued to $\mathbb{O}^*$ with exactly one pole; it is simple, lies at a real point, and its value there is the point at infinity $\infty = N$.
- (C2) *Infinite Euler product.* On a slice right half-space $\Omega_0 \subset \mathbb{O}^*$, $A = \exp\big(\sum_p \ell_p\big)$ for an *infinite* family $\{\ell_p\}$ of slice-preserving slice-regular functions each zero-free on Ω_0, the family summable *locally normally* on Ω_0 (the sense in which the cited Euler products converge; Theorem 1.2); in particular A is zero-free on Ω_0.
- (C3) *Infinite Weierstrass factorization.* Over the *full divisor* of A, including the single pole of (C1), at the real point p_0, multiplicity -1, carried by the explicit left factor, on $\mathbb{O}^* \setminus \{p_0\}$,

$$(q - p_0)\,A \;=\; q^m\,R\,e^g \prod_n \mathcal{E}(\,\cdot\,; q_n),$$

where q is the octonionic coordinate, $m \geq 0$, R is the slice-regular Weierstrass product over the *nonzero* residue-$\mathbb{R}$ (real) zeros of A, vanishing only on $\mathbb{R}$, g is slice-preserving entire, $\{q_n\}$ enumerates the residue-$\mathbb{C}$ zero-spheres of A, and $\mathcal{E}(\,\cdot\,; q_n)$ is the slice-regular Weierstrass primary factor of q_n (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite, the product converging *locally normally* on $\mathbb{O}^* \setminus \{p_0\}$ (the convergence of Proposition 10.1). (The left factor makes "full divisor" literal: the divisor of a meromorphic section includes its single pole; classically, Hadamard factors $(s-1)\zeta(s)$.)

(C4) *Infinitely many residue-$\mathbb{C}$ zeros.* The index set $\{q_n\}$ is infinite. A closed zero of such a section is a real point (residue field $\mathbb{R}$) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$; [2]); residue-$\mathbb{C}$ names the classically nontrivial zeros (Part 2).

Definition 10.5 (The residue subdiagram and its inclusion)**.** The *residue locus* of an A-section is $\{z : A(z) = 0,\ \operatorname{Im} z > 0\}$ (`CResidueZeroLocus`). The *residue subdiagram* $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$ (`AsectionCResidueDiagram`) is the preimage of the states this locus selects at the north frame — a preimage taken in the total action groupoid, not of a set — with the witnessing base arrow carried as data; its inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$ (`AsectionCResidueInclusion`) is a faithful embedding onto its image, and its naturality squares commute definitionally.

Theorem 10.6 (Concentricity)**.** *Let A be an A-section. The infinitely many C-residue zero S^6 spheres transported by the A-section functor $\mathcal{A}_A : \mathcal{B} \to$ **Grpd**, from the projective great-circle base $\mathcal{B}$ to the A-generated normalized value groupoids, form exactly one colimit/component class, and that singleton class carries one real value*

$$c \in \mathbb{R}.$$

Consequently every populated residue-$\mathbb{C}$ zero 6-sphere has centre c, so the infinitely many populated residue-$\mathbb{C}$ zero spheres are concentric. The theorem does not prescribe the numerical value of c.

Proof. We consider a slice-preserving section A of the ring $\mathcal{R}$ on the compactified octonions $\mathbb{O}^*$. Slice preservation hands us, for each slice, an isomorphism onto a Riemann sphere, pinned down by a normalization of $\mathbb{O}^*$; and this is already the push toward a functorial point of view, because the spheres do not come one at a time — they come as a family with coherent identifications, and such a family is organized by a functor. The hypotheses push in the same direction. Specifying the analytic object, the theory suggests exponential equations; and $\mathbb{O}^*$ comes from the Cayley–Dickson construction with only one real axis. The projective great circle that axis compactifies to is framed by $PGL(2,\mathbb{R})$: a projective coordinate frame in which $\exp(I\theta)$ lives, and the analytic sections of $\mathcal{R}$ themselves transform as Möbius transformations — self-transformations of the slice-preserving section. Frames and transformations compose, invert, and act: they are groupoids. That is the base $\mathcal{B}$ of Definition 9.4 (`GreatCircle.Base`), arrived at by the shape of the ring rather than by choice.

Conditions C1–C4 now determine what kind of distinguished action the A-section is. The exponential equations lead to the commuting triangle $\pi \circ E = \exp$ of the logarithm manifold (Theorems 6.1 and 6.2), giving the unique tame lift of any exponential base (Theorem 6.5, Proposition 6.6); since C2 supplies an Euler product, the winding of that lift is partitioned and well-defined at every prime, all of it carried in one diagonal Möbius element — a group and a function, and a functor. C3 supplies the unique factorization at N (Proposition 10.1). That the Euler diagonal element and the Weierstrass factorization are connected by the same unique tame lift is what gives the distinguished disk action its structure: the degenerate fibre of Lemma 6.3, with its one real level and its winding heights, is this action's own level-and-height bookkeeping. Each of these points of view is by itself fragmentary; conjugated, they show how A-section functors act. And they act on groupoids, so the construction is completed by categorified orbit–stabilizer theory, which is how such a functor is made well-defined on all of $\mathbb{O}^*$: the Möbius group law gives the local action, orbit representatives position the element at every footpoint, stabilizer compatibility makes the positioning independent of the representatives, and the group identity and multiplication supply the functor laws. The G_2-equivariant realization (Theorem 7.4) lifts the same action to the value states, along the continuum of Riemann spheres through the one point N. The result is the A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$ of Definition 9.4, whose states carry their real-value data and whose arrows are the element's actual value transports.

From the functor we form its total Grothendieck construction, $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ (Definition 9.4): an object is a base position together with a normalized value state in its fibre, a morphism is a base channel together with the compatible transport, and the whole is an action groupoid formed from the distinguished element. It is in this total, and not in any set, that the residue-$\mathbb{C}$ zeros are found. The distinguished element fixes 0 and N, and between them the upper half-space; at the north frame, where the positioning factor is trivial and the frame is the element itself, the equation $A = 0$ on that half-space selects the residue states, with Euler presenting at one end, Weierstrass at the other. The preimage of this selection under the action — the preimage of the total action groupoid, membership travelling by composition in the base — is the residue subdiagram $\mathcal{R}_A$ of Definition 10.5, included in $\mathcal{A}_A$ by ι_A. The restriction of the total object to this preimage sits over the same base.

Connectedness is established here, by the structure of ι_A itself. Because ι_A is a faithful embedding, an isomorphism onto its image, there is a canonical diagram of three categories: the domain $\mathcal{R}_A$, the image subcategory $\mathrm{Im}(\iota_A)$, and the full target $\mathcal{A}_A$. This creates a mandatory span of functors, a length-two zigzag:

$$\begin{tikzcd} & \mathrm{Im}(\iota_A) \arrow[dl, "\iota_A^{-1}"'] \arrow[dr, hook, "j"] & \\ \mathcal{R}_A & & \mathcal{A}_A \end{tikzcd}$$

This follows immediately, because ι_A is a *proper* inclusion and a natural isomorphism onto its image. The backward path exists because $\iota_A : \mathcal{R}_A \to \mathrm{Im}(\iota_A)$ is a strict isomorphism of categories, hence fully invertible; the forward path is the canonical, non-invertible inclusion $j : \mathrm{Im}(\iota_A) \hookrightarrow \mathcal{A}_A$. Any relation of the full category back to the restricted domain traverses exactly this roof.

It is worth pausing over what an arrow in this total *is*, because the conclusion turns on the answer. In an action groupoid, a morphism between two states is not a path, and not a chain of comparisons: it is a single element of the acting group, carrying one state to the other. The Lean library states this as an identity of types: a hom $p \to q$ *is* $\{m : M \mid m \cdot p = q\}$ (`CategoryTheory.ActionCategory.hom_as_subtype`). In a Grothendieck construction, an arrow *is* a pair — a base channel together with the compatible fibre transport over it (`CategoryTheory.Grothendieck.Hom`). These are two spellings of one movement: read on the total, an arrow of $\int_{\mathcal{B}} \mathcal{R}_A$ is one element of the distinguished action, presented by its base part and its value-transport part.

$$\begin{tikzcd} (X,\, x) \arrow[r, bend left, "{(f, \varphi_f)}"] \arrow[r, bend right, "m"'] & (Y,\, y) \end{tikzcd} \qquad \text{one arrow, two spellings}$$

So to join two residue states is to exhibit one element of the action that carries one to the other — and the action reaches every member of the preimage by definition rather than by derivation. The preimage is the C-residue locus: each member is by definition a state on one of the infinitely many S^6 C-residue zero spheres, so the inclusion lands in exactly the image on which the G_2-action is already transitive (Theorem 7.5). Orbit representatives position the element at every footpoint of the base, membership travels by composition, and the residue states form a single orbit under the action that produced them. A category is connected when its objects are joined by finite zigzags of morphisms; in an action groupoid the zigzag required has length one. The preimage is populated by infinitely many S^6 C-residue zero-sphere states (C4) and single-orbit — and that system is still an action groupoid. Taken together: ι_A is a fully faithful, transitive inclusion, and a fully faithful transitive inclusion of an action groupoid is connected — this is the connectedness instance of the Lean library's `ActionCategory` itself, transported along ι_A's proper inclusion: $\int_{\mathcal{B}} \mathcal{R}_A$ is a connected action groupoid. The element's unique tame continuous lift is what runs the members into one another while fixing the real level, a difference of winding being purely vertical (Lemma 6.3).

By Remark 8.3.5 of [23], a category is nonempty and connected if and only if its π_0 is the singleton set. Condition C4 supplies nonemptiness — infinitely many residue-$\mathbb{C}$ zero-sphere states populate the preimage — and connectedness was just established. Therefore, by Lemma 10.3 (`pi0GrothendieckEquiv`),

$$\pi_0\Big(\int_{\mathcal{B}} \mathcal{R}_A\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}} \,(\pi_0 \circ \mathcal{R}_A)$$

collapses to a singleton κ. The class and its value are kept in distinct registers: κ is the unique transport class, and $c \in \mathbb{R}$ is the one real level present in the states κ identifies, conserved along every connecting transport by the lift's level law; the level read is `ASection.transportLevel` in the Lean source. Reading the level on κ returns a single number. The populated residue-$\mathbb{C}$ zero 6-sphere states all represent κ, so each has centre c. Hence, the infinitely many residue-$\mathbb{C}$ zero spheres of the A-section are concentric. □

Corollary 10.7 (Translation to the classical framework)**.** *Under the residue-$\mathbb{C}$/classically-nontrivial-zero identification, the already-proved populated residue-$\mathbb{C}$ zero spheres are the classically nontrivial zero spheres and retain the common centre c supplied by Theorem 10.6.*

Proof. This is the translation of the C-residue population through Theorems 8.1 and 8.2. It neither constructs the centre nor proves a second concentricity statement; infinitude and disjointness are C4 and Lemma 10.2. □

11 Corollaries

Two corollaries close the argument. The first checks that the octonionic zeta is an A-section, verifying its four conditions C1–C4 one by one against the classical facts of Part 1; the Concentricity Theorem then applies to it directly, and its residue-$\mathbb{C}$ zero-spheres share one real centre c. The second corollary is the Riemann Hypothesis. The functional equation of ξ now pins that centre: it sends a zero of real part c to one of real part $1-c$, and applied to the common centre it forces $c = 1 - c$, so $c = \frac{1}{2}$. Every nontrivial zero lies on the critical line.

Corollary 11.1 ($\zeta_{\mathbb{O}^*}$ is an A-section)**.** *$\zeta_{\mathbb{O}^*}$ is an A-section* (Definition 10.4)*: a section of $\mathcal{R}$ satisfying* (C1)–(C4).

Proof. $\zeta_{\mathbb{O}^*} \in \mathcal{R}$ by slice-preservation (Theorem 7.2). The verification of (C1)–(C4) cites the classical facts of the Riemann zeta function (Part 1) and the slice-regular zero-geometry that translates them:

- **(C1)** *Meromorphic continuation; simple pole.* Classical fact: ζ is meromorphically continued with a single simple pole at $s = 1$ (Theorem 1.1); the value of $\zeta_{\mathbb{O}^*}$ there is the point at infinity, $\zeta_{\mathbb{O}^*}(1) = \infty = N$ (Definition 4.1).

- **(C2)** *Infinite Euler product.* Classical fact: $\zeta = \prod_p (1 - p^{-s})^{-1} = \exp(\sum_p \ell_p)$ on $\Re s > 1$, the family indexed by the infinitely many primes (Theorem 1.2).

- **(C3)** *Infinite Weierstrass factorization.* Supplied by the *slice-regular* Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]): $(q-1)\,\zeta_{\mathbb{O}^*} = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n)$ (pole factor at $p_0 = 1$; classically, Hadamard factors $(s-1)\zeta(s)$), where $m = 0$ ($\zeta(0) = -\frac{1}{2} \neq 0$), R is the product over the nonzero residue-$\mathbb{R}$ zeros, the classically trivial zeros $-2, -4, \ldots$, honest real zeros of $\zeta_{\mathbb{O}^*}$, and $\{q_n\}$ enumerates its residue-$\mathbb{C}$ zero-spheres.

- **(C4)** *Infinitely many residue-$\mathbb{C}$ zeros.* Classical fact: ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). By the residue dictionary below these are exactly the residue-$\mathbb{C}$ zeros, so $\{q_n\}$ is infinite.

The one remaining classical fact enters the downstream centre-pinning rather than (C1)–(C4): the *functional equation* $\xi(s) = \xi(1-s)$, $\xi(\bar{s}) = \overline{\xi(s)}$ (Theorem 1.1), used in Corollary 11.2. The *residue dictionary*, a closed zero of $\zeta_{\mathbb{O}^*}$ is a real point (residue field $\mathbb{R}$, the classically *trivial* zeros) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$, the classically *nontrivial* zeros), is the slice-regular zero-geometry of Theorem 8.2 and [1]; it is what makes "residue-$\mathbb{C}$" and "nontrivial" the same notion. Thus (C1)–(C4) hold. □

Corollary 11.2 (Riemann Hypothesis)**.** *Every nontrivial zero of the classical Riemann zeta function has real part $\frac{1}{2}$; in particular infinitely many zeros lie on the line $\Re s = \frac{1}{2}$.*

Proof. By Corollary 11.1, $\zeta_{\mathbb{O}^*}$ satisfies (C1)–(C4). By Theorem 10.6, its populated residue-$\mathbb{C}$ value-transport colimit is the real singleton $\{c\}$ and its residue-$\mathbb{C}$ zero spheres are already concentric about c. By Corollary 10.7, these are the classically nontrivial zeros, so every nontrivial zero has real part c. The functional equation $\xi(s) = \xi(1-s)$ (Theorem 1.1) sends a zero of real part c to one of real part $1-c$; applied to the *common* centre it forces $c = 1 - c$, whence $c = \frac{1}{2}$ (this is Theorem 8.3 read on $\zeta_{\mathbb{O}^*}$). Therefore every nontrivial zero has real part $\frac{1}{2}$, and by (C4) infinitely many lie on $\Re s = \frac{1}{2}$. □

A word on why the argument takes the shape it does. No one can hold an 8-dimensional graph in the mind's eye, or draw the continuum of Riemann spheres the section lives on. The epigraph names the way through: no single viewpoint sees the whole, but when the viewpoints are multiplied and conjugated, a vision appears that none of them held alone. Each slice direction is one such viewpoint; the value-bearing A-section functor gathers all of them over the base, the total object holds them together, and the colimit conjugates them into a single answer. The concentricity we could not picture becomes one component we can compute.

Remark 11.3 (Two alternative routes)*.* Two other proofs of Theorem 10.6 are available. The first runs over the base. The restriction sits over $\mathcal{B}$ through its two projections,

$$\begin{tikzcd} \int_{\mathcal{B}} \mathcal{R}_A \arrow[rr, hook, "\int \iota_A"] \arrow[dr, "\pi_{\mathcal{R}}"'] & & \int_{\mathcal{B}} \mathcal{A}_A \arrow[dl, "\pi_{\mathcal{A}}"] \\ & \mathcal{B} & \end{tikzcd}$$

and one may connect residue states footpoint by footpoint: zigzags in the connected base, lifted through the fibres. That zigzag is a different one from the single arrow used above — it travels through $\mathcal{B}$ and reassembles the value data leg by leg — and it is not the proof given here. The second is Thomason's homotopy colimit theorem [25]: the classifying space of a Grothendieck construction is homotopy equivalent to the homotopy colimit of the classifying spaces of its fibres, and reading connected components through that equivalence yields the same collapse of $\pi_0\big(\int_{\mathcal{B}} \mathcal{R}_A\big)$. The proof given here stays at the level of categories, where Lemma 10.3 suffices: the base route assembles what the action already supplies whole, the Thomason route is the topological shadow of the same computation, and neither path has been formalized in Lean.

References

[1] A. Altavilla and C. De Fabritiis, *$*$-logarithm for slice regular functions*, Rend. Lincei Mat. Appl. (arXiv:2106.04227); the slice-regular exponential $\exp_*$ (Def. 2.14, and Rem. 2.23 for the slice-preserving formula $\exp_*(f) = e^{f_0}(\cos f_1 + I \sin f_1))$ and the $*$-logarithm on never-vanishing functions (Def. 2.24, Thm. 6.17).

[2] A. Altavilla and C. De Fabritiis, *s-regular functions which preserve a complex slice*, Ann. Mat. Pura Appl. (arXiv:1801.01318); Prop. 3.1 and Thm. 3.2 factor a slice-regular function into a factor carrying exactly its (real and spherical) zeros and a never-vanishing factor, including the case of infinitely many zeros, the citable form of C3.

[3] J. C. Baez, *The octonions*, Bull. Amer. Math. Soc. **39** (2002), 145–205.

[4] C. Bisi and J. Winkelmann, *Invariants and Automorphisms for slice regular functions: the octonionic case*, arXiv:2411.16762 (2024); companion arXiv:2411.15896 in J. Noncommutative Geometry (EMS). Supplies the I-independent lift $\phi_I \circ F = A \circ \phi_I$ (the slice projection) and, in §3.2/§3.7/§14, the axially-symmetric-domain, slice-preserving, and central-divisor notions.

[5] F. Colombo, I. Sabadini, and D. C. Struppa, *Noncommutative Functional Calculus: Theory and Applications of Slice Hyperholomorphic Functions*, Progress in Mathematics **289**, Birkhäuser, 2011.

[6] R. Ghiloni and A. Perotti, *Slice regular functions on real alternative algebras*, Adv. Math. **226** (2011), 1662–1691.

[7] R. Ghiloni, A. Perotti, and C. Stoppato, *Singularities of slice regular functions over real alternative $*$-algebras*, Adv. Math. **305** (2017), 1085–1130 (slice-function and stem-function formalism, §1–§2; the slice-preserving $\Leftrightarrow$ intrinsic $\Leftrightarrow$ $\mathbb{R}$-valued-stem characterization used in Definition 6.4; §11: slice semiregular functions, Def. 11.1, Thm. 11.3).

[8] G. Gentili and D. C. Struppa, *A new theory of regular functions of a quaternionic variable*, Adv. Math. **216** (2007), 279–301.

[9] G. Gentili, C. Stoppato, and D. C. Struppa, *Regular Functions of a Quaternionic Variable*, Springer Monographs in Mathematics, Springer, 2013.

[10] G. Gentili and I. Vignozzi, *The Weierstrass factorization theorem for slice regular functions over the quaternions*, Ann. Global Anal. Geom. **40** (2011), 435–466.

[11] G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, C. R. Acad. Sci. Paris **158** (1914), 1012–1014.

[12] J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000.

[13] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berlin. Akad. (1859).

[14] R. D. Schafer, *An Introduction to Nonassociative Algebras*, Academic Press, 1966.

[15] E. C. Titchmarsh (rev. D. R. Heath-Brown), *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.

[16] X. Wang, *On geometric aspects of quaternionic and octonionic slice regular functions*, arXiv:1512.01414v3.

[17] G. Gentili, J. Prezelj, and F. Vlacci, *Slice conformality and Riemann manifolds on quaternions and octonions*, Math. Z. **302** (2022), 971–994 (arXiv:2107.07892v2).

[18] The Stacks Project Authors, *The Stacks Project*, https://stacks.math.columbia.edu, 2024.

[19] The mathlib Community, *The Lean mathematical library* (`CategoryTheory.ActionCategory`, `CategoryTheory.ConnectedComponents`, `CategoryTheory.Comma`), https://leanprover-community.github.io/mathlib4_docs/.

[20] P. G. Goerss and J. F. Jardine, *Simplicial Homotopy Theory*, Modern Birkhäuser Classics, Birkhäuser, 2009 (Ch. I: simplicial sets and the nerve; Ch. IV: the bisimplicial engine for homotopy colimits).

[21] G. Gentili, J. Prezelj, and F. Vlacci, *On a continuation of quaternionic and octonionic logarithm along curves and the winding number*, J. Math. Anal. Appl. **536** (2024), no. 1, Paper No. 128219, DOI 10.1016/j.jmaa.2024.128219 (arXiv:2307.14047) (Rem. 2.1: the direction $I(q)$ has no continuous extension to $\mathbb{R}$; Def. 4.7: tame path = unique companion (Def. 4.20 for maps; Def. 5.2: tame/semi-tame at an obstruction parameter); Def. 5.11: the loop lift $\mathrm{pr}_1\circ\Gamma = \gamma\circ\exp$; Cor. 5.13: a lift of a loop exists iff the signature lies in $\{0, -1\}$ on each obstruction interval, and is then a loop; Cor. 5.21: winding $= |\sigma^c|/2$).

[22] D. Quillen, *Higher algebraic K-theory: I*, in *Algebraic K-theory, I*, Lecture Notes in Math. **341**, Springer, 1973, 85–147 (Theorem A: a functor whose comma categories are contractible is final and induces an equivalence of classifying spaces).

[23] E. Riehl, *Categorical Homotopy Theory*, New Mathematical Monographs **24**, Cambridge University Press, 2014 (the nerve, classifying spaces, the two-sided bar construction, and homotopy colimits; Ch. 1, 4–6).

[24] E. Riehl, *Category Theory in Context*, Aurora: Dover Modern Math Originals, Dover Publications, 2016 (action groupoids and their orbits; Ex. 1.5.19).

[25] R. W. Thomason, *Homotopy colimits in the category of small categories*, Math. Proc. Cambridge Philos. Soc. **85** (1979), 91–109 (Thm. 1.2: the classifying space of the Grothendieck construction as a homotopy colimit).

[26] A. Vistoli, *Notes on Grothendieck topologies, fibered categories and descent theory*, in *Fundamental Algebraic Geometry*, Math. Surveys Monogr. **123**, AMS, 2005 (arXiv:math/0412512); §3.1 for the Grothendieck construction and categories fibered in groupoids.